

University of East London

School of Architecture, Computing and Engineering

Lab Report

CN6005 Artificial Intelligence

Module Tutors: Dr. Maimoona Sharif

Module Leader: Dr. Nadeem Qazi

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Week 5: KMEAN vs TFID & BERT Embedding on Diabetes Dataset

Student ID: 2923566

Student Name: Mazen Kozbar

- **Comparison:**

To see how much the choice of representation affects clustering, the same K-Means algorithm (two clusters, matching the diabetic and non-diabetic split) was run on two versions of the diabetes data. Each patient record was first written out as a short text description. In the first version, that text was converted with TF-IDF, giving 100 features. In the second, it was converted into Sentence-BERT embeddings and reduced with PCA to 50 dimensions. Both were judged by how cleanly the clusters separate, using three internal measures that need no diagnosis labels.

Embedding Method	Feature Dimensions	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Score
TF-IDF Vectorizer	100	0.017	4.843	16.90
Sentence-BERT + PCA	50	0.202	1.762	226.86

Sentence-BERT came out ahead on every measure, and it did so with half as many features. Its silhouette score of 0.202 is roughly twelve times TF-IDF's 0.017. The Davies-Bouldin index fell from 4.84 to 1.76, which means its clusters overlap far less. The Calinski-Harabasz score rose from 16.9 to 226.9, which means the groups are tighter inside and further apart from each other.

The most likely reason is how each method treats numbers. TF-IDF only counts which tokens appear, so a glucose reading of 148 and one of 150 are as different to it as any two unrelated words. Sentence-BERT builds a dense representation in which similar descriptions end up close together, and PCA then removes noise before the clustering step. That gives K-Means a much cleaner structure to work with.

These results should still be read with some caution. A silhouette score of 0.202 is an improvement, but it still points to only moderate separation, so the clusters are far from perfectly distinct. The two methods also work in different feature spaces, which means the scores are a useful guide rather than an exact like-for-like measurement. Finally, none of these measures checks the clusters against the real diagnoses. The sensible next step is to compare each cluster with the Outcome column, to confirm that the better separation reflects genuine diabetic and non-diabetic groups.